

Student Living Cost Analysis

Problem overview

The HSGA is looking to push for stipend increase within the various College of Medicine departments. They need to provide substantial and convincing information to the Office of Graduate Education (OGE) to inform this adjustments.

Data source

91 students got to take part in a google forms survey between **2025/08/20 11:02:56 AM AST - 2025/09/05 10:41:37 AM AST** . The aim of the survey was to get a glimpse into students cost of living. Data was then uploaded as a csv file.

Methodology

1. Loading data and preview
2. Exploratory Data Analysis
3. Conclusion, with recommendations

```
In [1]: #Importing necessary libraries to load and manipulate our data  
import pandas as pd  
import sqlite3
```

```
In [2]: #Loading our data into a DataFrame  
df = pd.read_csv('Data/Student Living Cost Survey 2025-2026.csv')
```

```
In [3]: df.rename(columns=lambda x: x.strip().replace(" ", "_").replac
```

We are going to read our data from SQLite DB for better presentation.

```
In [4]: #Creating a connection with our Survey DB  
conn = sqlite3.connect('Data/Survey.db')
```

```
In [5]: #Writing df into sqlite table  
df.to_sql("Survey", conn, if_exists="replace", index=False)
```

```
In [6]: #Selecting all columns in the Survey table  
query = """  
        SELECT *  
        FROM Survey;  
        """
```

```
In [7]: #Displaying our data
pd.read_sql(query, conn)
```

Out[7]:

	Timestamp	What_is_your_program	Are_you_an_international_student	Wt
0	2025/08/20 11:02:56 AM AST	Molecular and Developmental Biology	Yes	
1	2025/08/20 11:02:58 AM AST	Biomedical Informatics/Biostats	Yes	
2	2025/08/20 11:05:59 AM AST	Molecular Genetics, Biochemistry and Microbiology	No	
3	2025/08/20 11:07:21 AM AST	Molecular and Developmental Biology	Yes	
4	2025/08/20 11:09:05 AM AST	Epidemiology	Yes	
...	...	...	...	
86	2025/09/04 7:36:16 AM AST	Molecular, Cellular and Biochemical Pharmacology	Yes	
87	2025/09/05 8:33:07 AM AST	Genetic Counseling	No	
88	2025/09/05 9:08:11 AM AST	Molecular and Developmental Biology	No	
89	2025/09/05 10:22:00 AM AST	Biomedical Informatics/Biostats	No	
90	2025/09/05 10:41:37 AM AST	Neuroscience	No	

91 rows × 18 columns



```
In [8]: #Selecting rows with international students
international = pd.read_sql("""
                        SELECT Are_you_an_international_student
                        FROM Survey
                        WHERE Are_you_an_international_student = 'Yes'
                        """, conn )
```

```
In [9]: #Checking number of international students
international.count()
```

```
Out[9]: Are_you_an_international_student    26
dtype: int64
```

```
In [10]: #Inspecting number of non-international students
non = pd.read_sql("""
                SELECT Are_you_an_international_student
                FROM Survey
                WHERE Are_you_an_international_student = 'No'
                """, conn )
non.count()
```

```
Out[10]: Are_you_an_international_student    65
dtype: int64
```

Great!! So out of 91 students, we have 65 non-international, making international 26

```
In [11]: df.columns
```

```
Out[11]: Index(['Timestamp', 'What_is_your_program', 'Are_you_an_inter
national_student',
               'What_was_your_individual_monthly_housing_payment_duri
ng_the_previous_year_(Base_cost;_no_utilities)',
               'What_is_your_individual_monthly_housing_payment_for_t
he_upcoming_year_(Base_cost;_no_utilities)',
               'What_is_your_monthly_cost_on_utilities_(gas/electric/
water/trash/internet/laundry_services/renter's_insurance,_et
c.)',
               'Do_you_consider_your_neighborhood_to_be_a_safe_enviro
nment_Elaborate_on_your_point.',
               'If_you_answered_"No"_in_the_previous_question,_consid
ering_the_perceived_level_of_safety_in_your_current_neighborh
ood,_what_would_be_the_expected_monthly_rental_cost_of_an_apa
rtment_in_a_location_deemed_secure',
               'On_average,_how_much_time_do_you_spend_one-way_commut
ing_on_a_daily_basis_Please_specify_walk,_drive,_and/or_publi
c_transportation.',
               'What_is_your_average_monthly_cost_to_commute_to_work_
Please_include_gas,_Metro_bus,_and/or_taxi/Uber_prices.',
               'What_is_your_average_yearly_expenditure_incurred_for_
travel_to_your_home_or_for_vacation_purposes_Please_include_g
as,_Metro_bus,_taxi/Uber,_and/or_airfare_prices).',
               'If_you_own_a_car,_what_is_your_average_yearly_cost_fo
r_car_maintenance/car_payments/car_insurance/parking_passes',
               'What_is_your_average_monthly_(not_weekly)_expenditure
_on_groceries_and_dining_out',
               'What_is_your_average_monthly_expense_for_health-assoc
iated_costs_(medications/copays/procedures/etc.)',
               'For_pet_owners,_what_is_the_average_monthly_expenditu
re_on_pet-related_costs',
               'If_applicable,_approximately_how_much_do_you_spend_on
_average_per_month_to_support_dependents_(children,_elderly_p
arents,_etc.)',
               'If_applicable,_what_are_your_average_miscellaneous_ex
penses_per_year_that_are_essential_but_were_not_mentioned_in_
the_above_categories_(e.g._Clothing_etc.)_Please_list_each_ex
pense_category_in_your_answer.',
               'In_the_past_12_months,_which_of_the_following_trade-o
ffs_have_you_made_to_meet_living_expenses_(Select_all_that_ap
ply.)'],
              dtype='object')
```

```
In [12]: #Checking for missing values  
df.isna().sum()
```

Out[12]: Timestamp

0

What_is_your_program

0

Are_you_an_international_student

0

What_was_your_individual_monthly_housing_payment_during_the_previous_year_(Base_cost;_no_utilities)

1

What_is_your_individual_monthly_housing_payment_for_the_upcoming_year_(Base_cost;_no_utilities)

0

What_is_your_monthly_cost_on_utilities_(gas/electric/water/trash/internet/laundry_services/renter's_insurance,_etc.)

0

Do_you_consider_your_neighborhood_to_be_a_safe_environment_Elaborate_on_your_point.

0

If_you_answered_"No"_in_the_previous_question,_considering_the_perceived_level_of_safety_in_your_current_neighborhood,_what_would_be_the_expected_monthly_rental_cost_of_an_apartment_in_a_location_deemed_secure 70

On_average,_how_much_time_do_you_spend_one-way_commuting_on_a_daily_basis_Please_specify_walk,_drive,_and/or_public_transportation.

1

What_is_your_average_monthly_cost_to_commute_to_work_Please_include_gas,_Metro_bus,_and/or_taxi/Uber_prices.

1

What_is_your_average_yearly_expenditure_incurred_for_travel_to_your_home_or_for_vacation_purposes_Please_include_gas,_Metro_bus,_taxi/Uber,_and/or_airfare_prices).

3

If_you_own_a_car,_what_is_your_average_yearly_cost_for_car_maintenance/car_payments/car_insurance/parking_passes

18

What_is_your_average_monthly_(not_weekly)_expenditure_on_groceries_and_dining_out

0

What_is_your_average_monthly_expense_for_health-associated_costs_(medications/copays/procedures/etc.)

0

For_pet_owners,_what_is_the_average_monthly_expenditure_on_pet-related_costs

50

If_applicable,_approximately_how_much_do_you_spend_on_average_per_month_to_support_dependents_(children,_elderly_parents,_

etc.)

57

If_applicable,_what_are_your_average_miscellaneous_expenses_per_year_that_are_essential_but_were_not_mentioned_in_the_above_categories_(e.g._Clothing_etc.)_Please_list_each_expense_category_in_your_answer. 7

In_the_past_12_months,_which_of_the_following_trade-offs_have_you_made_to_meet_living_expenses_(Select_all_that_apply.)

27

dtype: int64

```
In [13]: # Converting relevant columns to numeric, coercing errors for
df['What_was_your_individual_monthly_housing_payment_during_the_past_12_months'] = df['What_was_your_individual_monthly_housing_payment_during_the_past_12_months'].astype(float)
df['What_is_your_individual_monthly_housing_payment_for_the_upcoming_year'] = df['What_is_your_individual_monthly_housing_payment_for_the_upcoming_year'].astype(float)
df['What_is_your_monthly_cost_on_utilities_(gas/electric/water)'] = df['What_is_your_monthly_cost_on_utilities_(gas/electric/water)'].astype(float)
df['What_is_your_average_monthly_(not_weekly)_expenditure_on_groceries'] = df['What_is_your_average_monthly_(not_weekly)_expenditure_on_groceries'].astype(float)
```



Exploratory Data Analysis

```
In [14]: # EDA: Summary statistics
eda_summary = df[[
    'What_was_your_individual_monthly_housing_payment_during_t
    'What_is_your_individual_monthly_housing_payment_for_the_u
    "What_is_your_monthly_cost_on_utilities_(gas/electric/wate
    'What_is_your_average_monthly_(not_weekly)_expenditure_on_
]].describe()

print("Summary statistics for key cost categories:")
print(eda_summary)
```

Summary statistics for key cost categories:

```
What_was_your_individual_monthly_housing_payment_during_the_previous_year_(Base_cost;no_utilities) \
count      65.000000
mean      1075.876923
std       459.606303
min        0.000000
25%       800.000000
50%       980.000000
75%      1350.000000
max      3000.000000
```

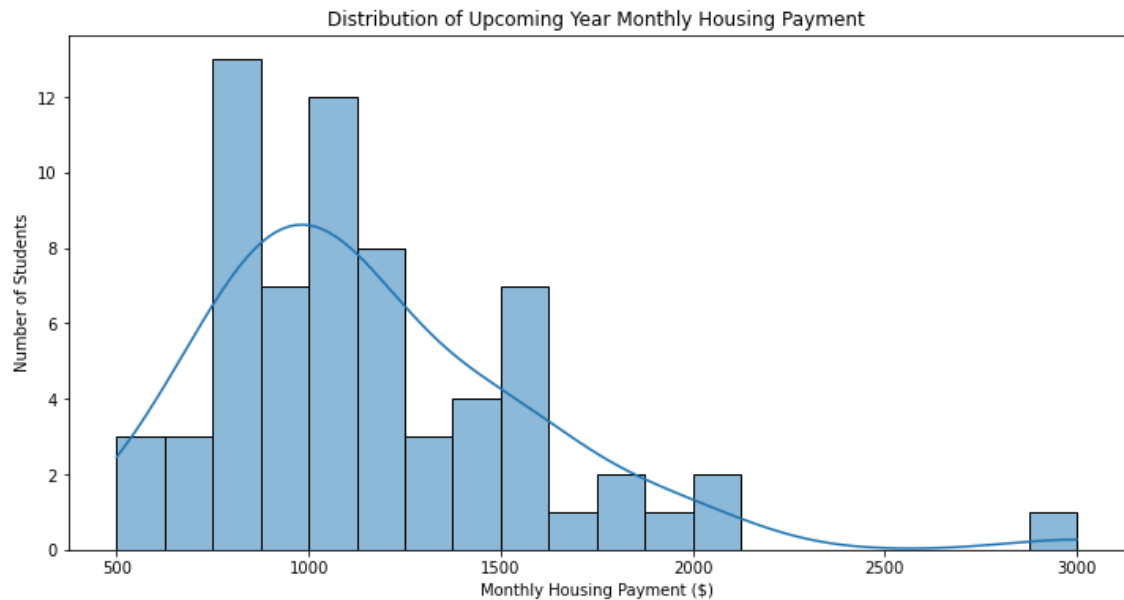
```
What_is_your_individual_monthly_housing_payment_for_the_upcoming_year_(Base_cost;no_utilities) \
count      67.000000
mean      1160.985075
std       433.219886
min       500.000000
25%       850.000000
50%      1085.000000
75%      1400.000000
max      3000.000000
```

```
What_is_your_monthly_cost_on_utilities_(gas/electric/water/trash/internet/laundry_services/renter's_insurance,_etc.) \
count      86.000000
mean      221.406977
std       116.943664
min       10.000000
25%       150.000000
50%       200.000000
75%       297.250000
max       700.000000
```

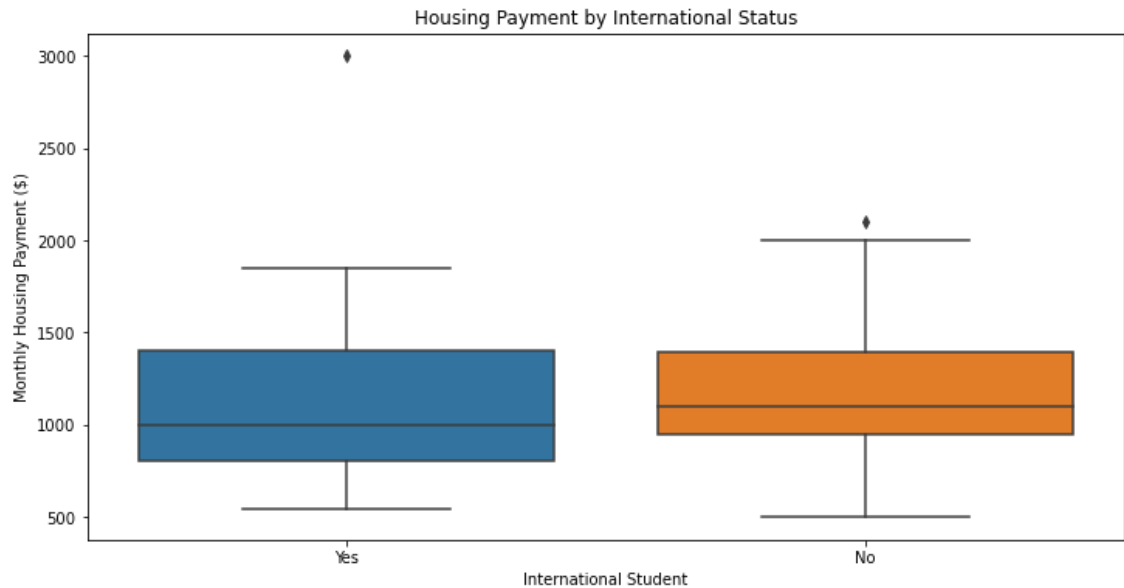
```
What_is_your_average_monthly_(not_weekly)_expenditure_on_groceries_and_dining_out
count      81.000000
mean      395.864198
std       286.555830
min        0.000000
25%       250.000000
50%       300.000000
75%       500.000000
max      2000.000000
```

```
In [15]: import seaborn as sns
import matplotlib.pyplot as plt

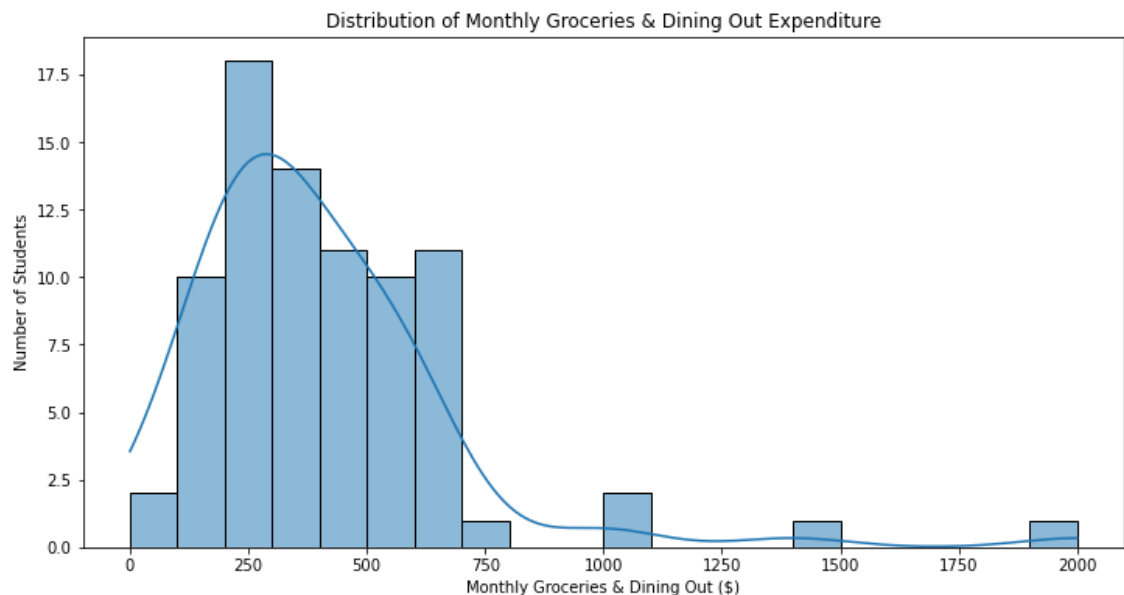
plt.figure(figsize=(12, 6))
sns.histplot(df['What_is_your_individual_monthly_housing_payme
plt.title('Distribution of Upcoming Year Monthly Housing Payme
plt.xlabel('Monthly Housing Payment ($)')
plt.ylabel('Number of Students')
plt.show()
```



```
In [16]: plt.figure(figsize=(12, 6))
sns.boxplot(data=df, x='Are_you_an_international_student',
            y='What_is_your_individual_monthly_housing_payment')
plt.title('Housing Payment by International Status')
plt.xlabel('International Student')
plt.ylabel('Monthly Housing Payment ($)')
plt.show()
```



```
In [17]: plt.figure(figsize=(12, 6))
sns.histplot(df['What_is_your_average_monthly_(not_weekly)_exp
plt.title('Distribution of Monthly Groceries & Dining Out Expe
plt.xlabel('Monthly Groceries & Dining Out ($)')
plt.ylabel('Number of Students')
plt.show()
```

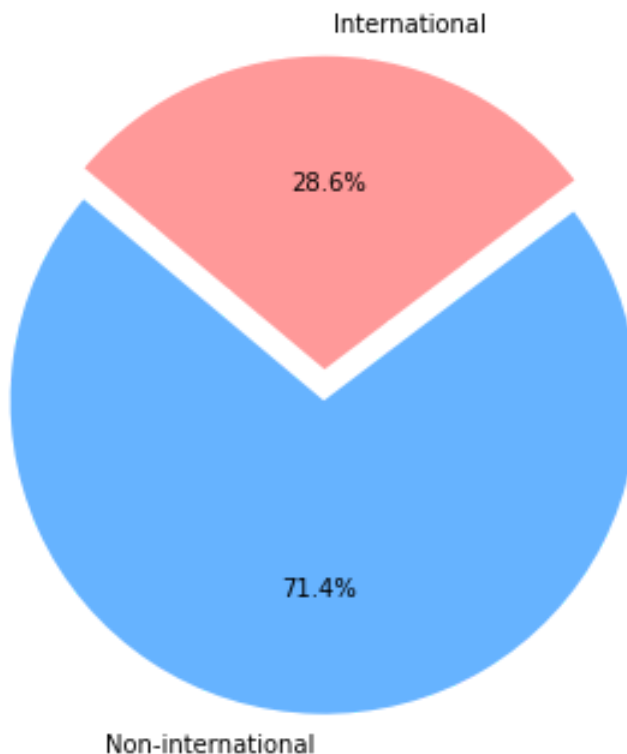


```
In [18]: from sklearn.linear_model import LinearRegression
import numpy as np

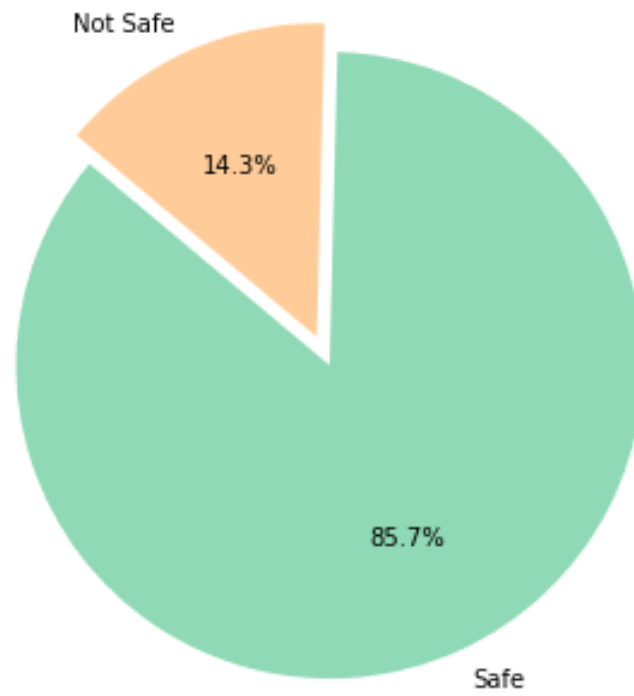
# Pie chart: International vs Non-international students
plt.figure(figsize=(6, 6))
labels = ['Non-international', 'International']
sizes = [len(non), len(international)]
colors = ['#66b3ff', '#ff9999']
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=14)
plt.title('International vs Non-international Students')
plt.show()

# Pie chart: Safe vs Not Safe Neighborhood
plt.figure(figsize=(6, 6))
safe_counts = df['Do_you_consider_your_neighborhood_to_be_a_sa
labels = ['Safe', 'Not Safe']
sizes = [safe_counts.get(True, 0), safe_counts.get(False, 0)]
plt.pie(sizes, labels=labels, autopct='%1.1f%%', startangle=14)
plt.title('Perceived Neighborhood Safety')
plt.show()
```

International vs Non-international Students



Perceived Neighborhood Safety



```
In [19]: # Analysis of the trade-offs students made to meet living expe

tradeoffs_col = 'In_the_past_12_months,_which_of_the_following

# Drop missing values and split multiple selections
tradeoffs = df[tradeoffs_col].dropna().str.get_dummies(sep=',')

# Sum up each trade-off option
tradeoff_counts = tradeoffs.sum().sort_values(ascending=False)

print("Trade-offs made by students to meet living expenses (co
print(tradeoff_counts)

# Plot the results
plt.figure(figsize=(10, 6), constrained_layout=True)
tradeoff_counts.plot(kind='bar', color="teal")
plt.title('Trade-offs Made by Students to Meet Living Expenses
plt.ylabel('Number of Students')
plt.xlabel('Trade-off')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
Trade-offs made by students to meet living expenses (count):
cheaper/less nutritious
41
medical
13
phone
13
skipped meals);Delayed medical/dental care or prescriptions due to cost;Used credit cards/loans or sold belongings to cover essentials
12
etc.);Reduced food quality/quantity (e.g.
12
Lived farther from campus to reduce housing/parking costs;
Reduced food quality/quantity (e.g.
9
skipped meals);Delayed medical/dental care or prescriptions due to cost
~
```

Performing Linear Regression so as to predict upcoming year housing payment from previous year payment

```

In [20]: # Prepare data, drop rows with missing values in relevant columns
reg_df = df[['What_was_your_individual_monthly_housing_payment_2022',
             'What_is_your_individual_monthly_housing_payment_for_2023']]

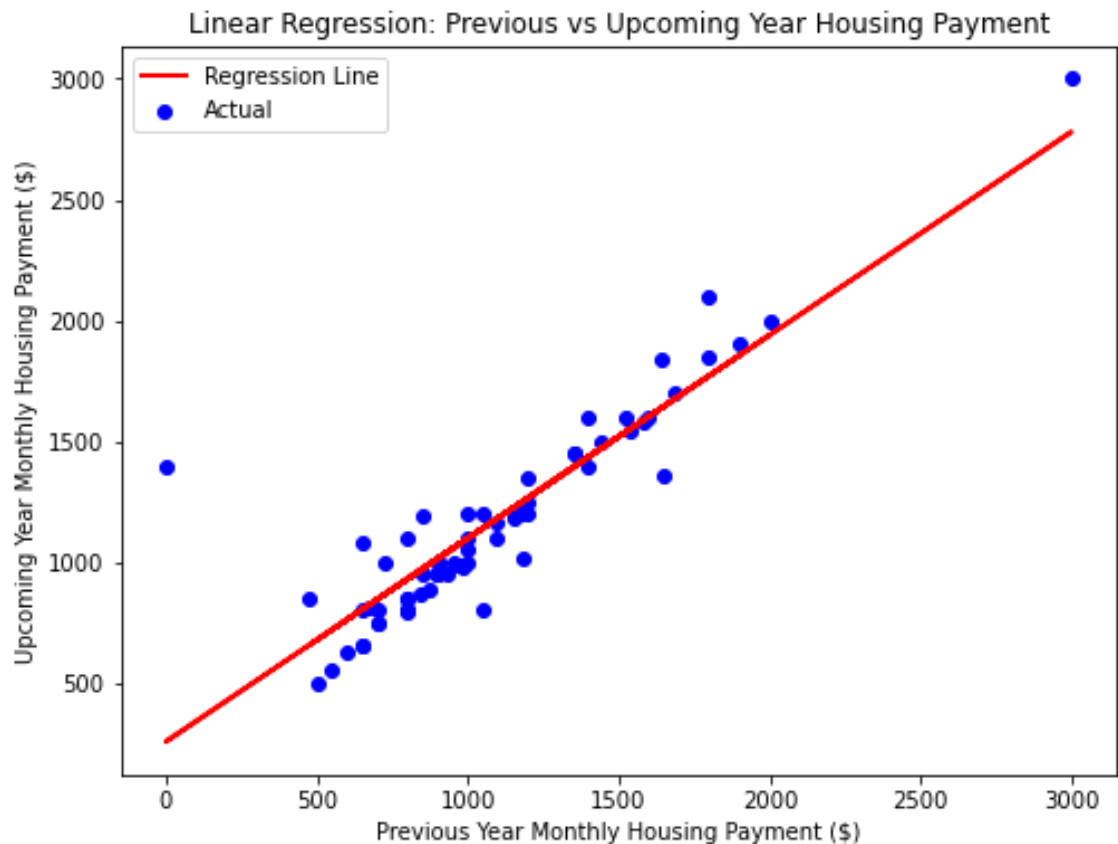
X = reg_df[['What_was_your_individual_monthly_housing_payment_2022']]
y = reg_df[['What_is_your_individual_monthly_housing_payment_for_2023']]

model = LinearRegression()
model.fit(X, y)
y_pred = model.predict(X)

plt.figure(figsize=(8, 6))
plt.scatter(X, y, color='blue', label='Actual')
plt.plot(X, y_pred, color='red', linewidth=2, label='Regression Line')
plt.xlabel('Previous Year Monthly Housing Payment ($)')
plt.ylabel('Upcoming Year Monthly Housing Payment ($)')
plt.title('Linear Regression: Previous vs Upcoming Year Housing Payment')
plt.legend()
plt.show()

print(f"Regression Coefficient: {model.coef_[0]:.2f}")
print(f"Regression Intercept: {model.intercept_:.2f}")
print(f"R^2 Score: {model.score(X, y):.2f}")

```



Regression Coefficient: 0.84

Regression Intercept: 258.46

R² Score: 0.80

The linear regression model predicts a student's upcoming year monthly housing payment based on their previous year's payment.

- The model uses the previous year's payment as the independent variable (X) and the upcoming year's payment as the dependent variable (y).
- After fitting the model, the regression coefficient indicates how much the upcoming payment is expected to change for each dollar change in the previous payment.
- The intercept represents the baseline upcoming payment when the previous payment is zero.
- The R² score shows how well the previous year's payment explains the variation in the upcoming year's payment (with 1.0 being a perfect fit).

In this analysis:

- A positive regression coefficient means indicates that students who paid more for housing last year are likely to pay more in the upcoming year as well.

Recommendations to improve student funding

1. Increase housing stipends : Median housing costs are high and rising. Consider raising the base stipend to match the median rent plus utilities.

2. Targeted support for international students : International students may face higher housing costs and fewer funding options. Provide additional grants or subsidies.

3. Food security programs : Some students report high monthly food expenses, preferring to go for cheaper, less nutritious foods. Implement or expand food assistance programs or campus food pantries.

4. Transportation subsidies : Offer discounted transit passes or parking subsidies, as commuting costs vary widely.

5 Emergency funds : Create flexible emergency funding for students